Finite-dimensional detection of reduced almost-periodic Banach cocycles

Matthew J. Colbrook

Department of Applied Mathematics and Theoretical Physics
University of Cambridge, United Kingdom
mjc249@maths.cam.ac.uk

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Abstract

We prove that every nonzero reduced cohomology class of an almost-periodic Banach representation is detected by an unbounded cocycle in a finite-dimensional quotient. Compact-group spectral approximation and separating finite-type quotient maps give the result for arbitrary topological groups.

Repository target. This manuscript concerns AIM problem 901 at revision `5d7e26e8444e`. The precise scope is stated below.

1 Main result and proof

Answer. The answer to the stated question is affirmative. In fact, local compactness of the acting group is unnecessary. The essential distinction in the proof is between a representation with finitely many irreducible types and a finite-dimensional representation: the former can have arbitrary Banach multiplicity spaces.

Theorem 1.1. *Let G be a topological group, let V be a real or complex Banach space, and let $\pi : G \to \mathrm{GL}(V)$ be a strongly continuous representation such that each orbit has compact norm closure. If a continuous 1-cocycle $b : G \to V$ does not belong to the compact-open closure of*

$$B^1(G, V) = \{g \mapsto v - \pi(g)v : v \in V\},$$

then there is a closed invariant subspace $W \subset V$ with $0 < \dim(V/W) < \infty$ such that $\sup_{g\in G} \|b(g) + W\| = \infty$.

1.1 The compact group associated with the representation

Uniform boundedness gives $M := \sup_{g\in G} \|\pi(g)\| < \infty$. Embed $\pi(G)$ into the product of the compact orbit closures, one factor for each $v \in V$. Every point in its product closure is a linear operator of norm at most M. Consequently its strong operator closure K is compact. Multiplication is jointly strongly continuous on uniformly bounded sets: if $A_\alpha \to A$ and $B_\alpha \to B$ strongly with $\sup_\alpha \|A_\alpha\| < \infty$, then

$$\|A_\alpha B_\alpha v - ABv\| \le \|A_\alpha\| \, \|B_\alpha v - Bv\| + \|(A_\alpha - A)Bv\| \longrightarrow 0.$$

If $\pi(g_\alpha) \to k$, compactness supplies a subnet along which $\pi(g_\alpha^{-1}) \to l \in K$; the displayed estimate implies $kl = lk = I$. Thus K is a compact group of invertible operators, with continuous inversion. Write ρ for its strongly continuous action on V. The image of G is dense in K.

1.2 Central spectral approximations

For now suppose V is complex. For an irreducible unitary representation τ of K, write $d_\tau = \dim \tau$, $\chi_\tau = \operatorname{tr} \tau$, and, with normalized Haar measure, define

$$E_{ij}^{\tau} = d_\tau \int_K \overline{\tau_{ij}(k)} \rho(k)\, dk, \qquad P_\tau = \sum_{i=1}^{d_\tau} E_{ii}^{\tau} = d_\tau \int_K \overline{\chi_\tau(k)} \rho(k)\, dk.$$

These formulas mean integration after applying the integrand to a vector. The resulting operators are bounded. Schur orthogonality and invariance of Haar measure give the usual matrix-unit relations and

$$P_\tau^2 = P_\tau, \qquad P_\tau P_\sigma = 0 \quad (\tau \neq \sigma), \qquad \rho(k) P_\tau = \sum_{i,j=1}^{d_\tau} \tau_{ij}(k) E_{ij}^{\tau}. \tag{901.1}$$

In particular, all P_τ commute with $\rho(K)$. On the range of any finite sum of these projections, the operator span of the representation is finite dimensional and the representation is continuous in operator norm. The range itself need not be finite dimensional.

There is a uniformly bounded net (R_α) of finite linear combinations of the P_τ such that $R_\alpha v \to v$ for every $v \in V$. Here is a construction which also addresses nonmetrizable K. Compact groups have a basis of conjugation-invariant neighborhoods of the identity. For each sufficiently small such neighborhood take a nonnegative continuous central function supported there, of Haar integral one. Peter–Weyl approximation [2, Theorem 1 and Proposition 6], followed by averaging under conjugation, approximates this function in $L^1(K)$ by a central trigonometric polynomial. Choose the approximation errors tending to zero and the L^1 norms at most two. Integration against ρ gives the asserted net: central trigonometric polynomials are finite sums of characters, concentration at the identity gives strong convergence, and $\|R_\alpha\| \leq 2M$. It follows, by a finite-net argument, that

$$\sup_{v \in C} \|R_\alpha v - v\| \longrightarrow 0 \quad \text{for every norm-compact } C \subset V. \tag{901.2}$$

Let $P_0 = \int_K \rho(k)\, dk$ be the projection onto the invariant vectors.

1.3 A finite-type cocycle lemma

Lemma 1.2. *Let U be the range of a finite sum of nontrivial isotypic projections. Suppose $c : G \to U$ is a continuous cocycle whose image in every finite-dimensional invariant quotient of U is bounded. Then c is an exact coboundary.*

Proof. First, such quotients separate points of U. Indeed, for $\ell \in U^*$,

$$D_\ell = \operatorname{span}\{\ell \circ \rho(k)|_U : k \in K\}$$

is finite dimensional by (901.1). Its common annihilator W_ℓ is closed and invariant, and U/W_ℓ is finite dimensional. If $u \neq 0$, choose ℓ with $\ell(u) \neq 0$.

The Haar average on U is zero. Since the representation on U is operator-norm continuous and has finite-dimensional operator span, this average can be approximated in operator norm by finite convex combinations of the operators $\pi(g)|_U$. Choose $g_1, \ldots, g_m \in G$ and $t_j \geq 0$, $\sum_j t_j = 1$, so that

$$A = \sum_{j=1}^{m} t_j \pi(g_j)|_U, \qquad \|A\| < 1, \qquad v = (I - A)^{-1} \sum_{j=1}^{m} t_j c(g_j). \tag{901.3}$$

For any finite-dimensional invariant quotient $q : U \to F$, the bounded cocycle qc is a coboundary. To see this directly, average an inner product on F over the compact linear group induced by K.

The affine action $z \mapsto \pi_F(g)z + qc(g)$ is then an action by Euclidean isometries with a bounded orbit. The unique circumcenter of that orbit is fixed. Thus, for some $w \in F$,

$$qc(g) = (I - \pi_F(g))w \qquad (g \in G).$$

On the quotient, $\|A_F\| \le \|A\| < 1$. Applying q to (901.3) and using this identity gives $qv = w$. Therefore

$$q(c(g) - (I - \pi(g))v) = 0 \qquad (g \in G).$$

The separating family of finite-dimensional quotients now gives $c(g) = (I - \pi(g))v$ for every g. □

Every quotient in this lemma is available as a quotient of V: if P_U is the sum of the corresponding isotypic projections, then $qP_U : V \to F$ is bounded, onto, and equivariant.

1.4 Proof of the theorem

We prove the contrapositive. Suppose that every finite-dimensional invariant quotient of V sends b to a bounded cocycle. On the trivial isotypic component, $P_0 b$ is a homomorphism into the additive group of V^K. Each scalar functional on this space, composed with P_0, is an equivariant quotient map. A bounded scalar homomorphism is zero, since $h(g^n) = nh(g)$ for every positive integer n. Hence $P_0 b = 0$.

For a nontrivial type τ, the cocycle $P_\tau b$ satisfies Lemma 1.2, because its finite-dimensional quotient maps extend to V. It is therefore an exact coboundary. Every $R_\alpha b$ is a finite linear combination of these coboundaries and is itself a coboundary. If $C \subset G$ is compact, continuity of b makes $b(C)$ norm compact. Equation (901.2) consequently gives

$$\sup_{g \in C} \|R_\alpha b(g) - b(g)\| \longrightarrow 0.$$

This proves the complex theorem with the required compact-open closure; no sequential-closure assumption has been made.

For real V, complexify V, π, and b. The complexified representation is still almost periodic: an orbit of $x + iy$ lies in the sum of the compact orbit closures of x and iy. If b were in the compact-open closure of complex coboundaries, taking real parts of their primitives would put it in the closure of real coboundaries. Thus a nonzero real reduced class remains nonzero after complexification. The complex result supplies a finite-dimensional complex equivariant quotient with unbounded image of b. Restrict that quotient map to V and take its finite-dimensional real image. Its kernel is the desired W. □

Scope and source. The statement resolves the catalogue's precise finite-quotient question, associated with Question 4.7 of Cornulier–Tessera [1]. The proof uses neither separability nor reflexivity of V, nor compact generation or local compactness of G. The uniform approximations have finite representation type, not finite rank; no approximation property of the underlying Banach space is assumed. The argument concerns reduced cohomology and unbounded quotient cocycles, as required in Problem 901.

Corollary 1.3 (Consequences for the source questions)**.** *Every locally compact group has property $\mathcal{AP}_{\mathrm{fd}}$ in the terminology of [1, Definition 4.1]. Consequently $\mathcal{WAP}_{\mathrm{ap}}$ and $\mathcal{WAP}_{\mathrm{fd}}$ are equivalent properties of locally compact groups.*

Proof. The first assertion is precisely Theorem 1.1. The second follows from [1, Proposition 4.3], which expresses $\mathcal{WAP}_{\mathrm{fd}}$ as the conjunction of $\mathcal{WAP}_{\mathrm{ap}}$ and $\mathcal{AP}_{\mathrm{fd}}$. Thus the same argument answers Questions 4.5–4.7 of that paper; these consequences are counted as one catalogue entry here. □

References

[1] Y. Cornulier and R. Tessera, *On the vanishing of reduced 1-cohomology for Banach representations*, Ann. Inst. Fourier **70** (2020), 1951–2003. doi:10.5802/aif.3363. Question 4.7 and Proposition 4.3.

[2] T. Tao, *The Peter–Weyl theorem, and non-abelian Fourier analysis on compact groups*, 23 January 2011, Theorem 1 and Proposition 6. https://terrytao.wordpress.com/2011/01/23/the-peter-weyl-theorem-and-non-abelian-fourier-analysis-on-compact-groups/. The group is assumed compact Hausdorff; normalized Haar measure makes the stated L^2 approximation sufficient for L^1 approximation.